N-way vs Recursive Bisection for VRA-Compliant Redistricting:
A Comprehensive 50-State Comparison

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Abstract

Graph partitioning algorithms offer two fundamental approaches to congressional redistricting: n-way partitioning (direct division into k districts) and recursive bisection (hierarchical binary splitting). This paper presents the first comprehensive empirical comparison of these methods for Voting Rights Act (VRA) compliance using edge-weighted graph partitioning across all 50 states.

We conduct ablation studies testing five weight factors (1×–100×) and four minority thresholds (40%–55%) for both methods, totaling 1,760 redistricting runs across 44 multi-district states. Our key finding challenges conventional assumptions: **both methods achieve statistically equivalent success rates** (n-way: 47.5%, recursive: 48.3%, $p = 0.634$), with negligible effect size (Cohen's $d = -0.018$).

However, methods exhibit distinct characteristics. N-way partitioning offers **67.5% faster runtime** (3.68s vs 11.33s average), making it preferable for rapid prototyping and large-scale analysis. Recursive bisection achieves a **higher performance ceiling** with optimal parameters (56.8% vs 52.3% best configuration), benefiting careful parameter tuning. Methods show different parameter sensitivities—n-way prefers lower thresholds (40%), recursive prefers higher thresholds (50%)—and state-specific advantages, with n-way winning in Virginia (+35%) and recursive in Connecticut (+45%).

This equivalence finding has important implications for redistricting reform: practitioners can confidently choose either method based on computational constraints (speed vs optimization) rather than effectiveness concerns. Both approaches produce comparable VRA compliance when properly configured with edge-weighting and total minority coalitions.

1 Introduction

Graph partitioning algorithms for congressional redistricting face a fundamental architectural choice: should states be divided directly into k districts (n-way partitioning) or hierarchically through repeated binary splits (recursive bisection)? While both approaches build on the same METIS optimization core [Karypis and Kumar, 1998], they differ in optimization structure, parameter sensitivity, and computational characteristics. This paper provides the first comprehensive empirical comparison of these methods for Voting Rights Act (VRA) compliance.

1.1 The Architectural Choice

N-way partitioning performs direct k-way division, partitioning the state graph into exactly k congressional districts in a single optimization. Edge weights influence all k districts simultaneously, creating a global optimization landscape where minority concentration in one district automatically affects all others. This "one-shot" approach offers computational efficiency but may struggle with complex constraint satisfaction.

Recursive bisection employs hierarchical decomposition, repeatedly splitting into two parts until reaching k districts. Each binary split is a simpler 2-way optimization, potentially more stable than complex k-way problems. However, early splitting decisions cascade through the tree—a suboptimal early split propagates to all descendants. Edge weights influence only local binary decisions, not global k-district structure.

These architectural differences suggest potential performance trade-offs. N-way's global view might better preserve minority communities across all districts but require longer computation. Recursive's local optimizations might execute faster but fragment minorities through cascading errors. Prior work has assumed these trade-offs are substantial, but no systematic comparison exists.

1.2 Voting Rights Act Compliance Challenge

The Voting Rights Act prohibits states from diluting minority voting strength through redistricting [Supreme Court, 1986]. Section 2 requires creation of majority-minority (MM) districts—where minority voters constitute $\geq 50\%$ of voting-age population—when feasible [Johnson, 2001]. Achieving VRA compliance algorithmically requires preserving geographically dispersed minority communities without explicit racial gerrymandering.

Edge-weighted graph partitioning offers a solution: assign higher weights $w_{ij} = \alpha$ to edges connecting minority-dense census tracts (minority percentage $\geq \tau$), making such edges "expensive" to cut. METIS minimizes total edge cut weight, thereby concentrating minority populations into contiguous districts. Two parameters control performance:

- **Weight factor** (α): Multiplier for minority-minority edges (tested: 1×, 5×, 10×, 50×, 100×)
- **Minority threshold** (τ): Percentage defining "minority tract" (tested: 40%, 45%, 50%, 55%)

Edge weights are:

$$w_{ij} = \begin{cases} \alpha & \text{if } m_i \geq \tau \text{ and } m_j \geq \tau \\ 1 & \text{otherwise} \end{cases} \tag{1}$$

where m_i is tract i's minority percentage (total non-White population).

The question: do n-way and recursive bisection respond differently to edge-weighting? Does architecture affect VRA performance?

1.3 Research Questions

This paper addresses four questions through comprehensive 50-state empirical analysis:

RQ1: Overall Performance — Do n-way and recursive bisection achieve equivalent VRA compliance rates, or does one method systematically outperform?

RQ2: Parameter Sensitivity — Do methods exhibit different optimal configurations? Does n-way require different weight factors or thresholds than recursive?

RQ3: State-Specific Behavior — Do certain state characteristics (minority percentage, geographic clustering, district count) favor one method?

RQ4: Speed-Quality Trade-offs — Does faster method (if any) sacrifice VRA performance? What practical implications guide method selection?

1.4 Preview of Key Findings

Our comprehensive ablation study (1,760 redistricting runs across 44 states, 20 configurations per method per state) yields surprising results:

Finding 1: Statistical Equivalence — Both methods achieve nearly identical success rates (n-way: 47.5%, recursive: 48.3%, difference: 0.8 percentage points). Paired t-test shows no significant difference ($t = -0.48$, $p = 0.634$), with negligible effect size (Cohen's $d = -0.018$). The architectural choice does not substantially impact VRA compliance.

Finding 2: Speed vs Optimization Trade-off — N-way executes 67.5% faster (3.68s vs 11.33s average). Recursive achieves higher performance ceiling with optimal parameters (56.8% vs 52.3% best configuration). Choice depends on priority: speed (n-way) or maximum tuning potential (recursive).

Finding 3: Different Parameter Landscapes — Methods prefer different configurations. N-way optimizes at lower thresholds (40%) with moderate weights (50×). Recursive optimizes at higher thresholds (50%) with heavy weights (100×). This suggests different optimization mechanisms despite shared METIS core.

Finding 4: State-Specific Advantages — Neither method dominates state-by-state. N-way wins 5 states (largest: Virginia +35%), recursive wins 8 states (largest: Connecticut +45%), 31 states tie. State characteristics (minority dispersion, urban clustering) interact with method architecture.

Finding 5: Bimodal Success Distribution — Both methods show bimodal state-level performance: states either succeed consistently (100% across configurations) or fail consistently (0% across configurations). Intermediate success rates are rare. This indicates VRA compliance depends primarily on state demographics/geography, not method choice.

1.5 Implications for Redistricting Practice

These findings transform method selection guidance. Previous work assumed architectural differences create substantial performance gaps, leading to complex decision frameworks. Our equivalence result simplifies: *both methods work equally well for VRA compliance when properly parameterized.*

Practical recommendation: Use n-way for rapid prototyping, ensemble generation, and exploratory analysis where speed matters. Use recursive for carefully tuned single plans where maximizing MM district count justifies longer computation. Both approaches are viable for production redistricting.

The different parameter sensitivities suggest practitioners should tune each method independently rather than assuming universal optimal configurations. N-way's preference for $\tau = 0.40$ and recursive's preference for $\tau = 0.50$ reflect underlying optimization dynamics—n-way benefits from broader minority definition, recursive from sharper minority boundaries.

1.6 Contributions

This paper makes four contributions:

(1) First Comprehensive Comparison — Systematic evaluation of n-way vs recursive bisection across all 50 states with extensive parameter ablation (1,760 runs). Establishes empirical equivalence with statistical rigor.

(2) Parameter Sensitivity Analysis — Identifies distinct optimal configurations for each method. Shows weight factor and threshold preferences differ by architecture, providing tuning guidance.

(3) State-Level Characterization — Maps which state characteristics predict method advantages. Identifies Virginia-type states (n-way better) and Connecticut-type states (recursive better).

(4) Practical Decision Framework — Provides speed-quality trade-off analysis guiding method selection based on computational constraints and performance requirements.

1.7 Paper Organization

Section 2 describes edge-weighted graph partitioning methodology, parameter configurations, and experimental design. Section 3 presents head-to-head comparison results: overall success rates, statistical significance, runtime analysis. Section 4 analyzes parameter sensitivity, identifying optimal configurations for each method. Section 5 examines state-by-state results, identifying geographic patterns and method-specific advantages. Section 6 discusses implications for redistricting practice and theoretical insights into optimization architectures. Section 7 concludes with recommendations for method selection.

2 Background: Graph Partitioning Methods

2.1 Voting Rights Act and Redistricting

The Voting Rights Act of 1965 prohibits jurisdictions from drawing electoral boundaries that dilute minority voting strength. Section 2, as amended in 1982, creates an obligation to avoid vote dilution through gerrymandering when three conditions are met (*Thornburg v. Gingles*, 1986) [Supreme Court, 1986]:

1. The minority group is sufficiently large and geographically compact to constitute a majority in a district
2. The minority group is politically cohesive

3. The white majority votes sufficiently as a bloc to usually defeat minority-preferred candidates

When these "Gingles preconditions" are satisfied, jurisdictions must create majority-minority (MM) districts where minority voters comprise $\geq 50\%$ of the voting-age population. Failure to do so constitutes unlawful vote dilution under Section 2.

2.2 Traditional Approaches to VRA Compliance

Redistricting mapmakers typically achieve VRA compliance through explicit racial gerrymandering: intentionally packing minority voters into supermajority districts (65-75% minority) to guarantee electoral victories [Grofman et al., 1992]. This "packing strategy" reflects three assumptions:

(1) Single-group analysis: Focus on largest minority group (e.g., "Black-majority" districts in Alabama, "Hispanic-majority" districts in California). Ignore potential coalitions across racial/ethnic lines.

(2) Supermajorities required: Target 65-75% minority to account for lower turnout, age demographics (children), and racially polarized voting. Supreme Court has recognized 65% as typical threshold for effective minority representation [Johnson, 2001].

(3) Intentional consideration of race: Mapmakers explicitly use racial data to draw boundaries, justified as necessary for VRA compliance despite tension with Equal Protection principles [Supreme Court, 1993].

Critics argue this approach itself constitutes racial gerrymandering violating *Shaw v. Reno* (1993) [Supreme Court, 1993]. Proponents counter that VRA compliance requires race-conscious mapmaking [Pildes and Niemi, 1997]. This tension between VRA mandates and Equal Protection constraints remains unresolved.

2.3 Coalition Districts: Legal Foundation

Bartlett v. Strickland (2009) [Supreme Court, 2009] addresses when coalition districts—where multiple minority groups together form a majority—can satisfy Section 2. The Court ruled:

Holding: Section 2 does not require creation of coalition districts where no single minority group exceeds 50% ("minority-majority" districts are required; "crossover" districts where minorities form plurality are not).

Recognition of coalitions: However, when total minority population (all groups combined) exceeds 50%, districts may be created under Section 2 if minorities demonstrate political cohesion as a coalition.

This ruling legitimizes algorithmic approaches that naturally create coalition districts through geographic optimization rather than single-group targeting.

2.4 Total Minority vs Single-Group Analysis

Traditional VRA analysis focuses on single racial groups:

- Alabama: 25.6% Black, 5.3% Hispanic, 1.5% Asian → focus on "Black-majority" districts

- California: 39.4% Hispanic, 5.4% Black, 15.5% Asian → focus on "Hispanic-majority" districts

This approach fails in diverse states where no single group reaches critical mass. Computing **total minority** (all non-White combined) reveals greater potential:

- Alabama: 36.9% total minority (25.6% + 5.3% + 1.5% + others)
- California: 65.3% total minority (39.4% + 5.4% + 15.5% + others)

Total minority calculation:

$$\text{minority}_i = \text{total\_pop}_i - \text{white\_non\_hispanic}_i \tag{2}$$

This encompasses Black, Hispanic/Latino, Asian, Native American, Pacific Islander, and multiracial populations. The key question is whether these diverse groups demonstrate political cohesion—the second *Gingles* precondition.

2.5 Algorithmic Redistricting Approaches

Recent work applies graph partitioning algorithms to redistricting [Duchin and Tenner, 2018, Fifield et al., 2020]. These methods offer structural immunity to partisan manipulation by operating without access to partisan data. However, VRA compliance remains challenging.

Compactness-only algorithms: Pure geographic optimization (minimizing perimeter-to-area ratios, edge cuts) tends to fragment minority communities across districts. If minorities are geographically dispersed, compactness pulls against concentration.

Multi-constraint optimization: Simultaneously balancing population and minority population as dual constraints. METIS supports multi-constraint partitioning with target weight distributions (tpwgts) [Karypis and Kumar, 1999]. However, tight population balance (0.5% tolerance) dominates loose minority balance (50% tolerance), preventing effective minority concentration.

Ensemble methods: Generate thousands of redistricting plans via Markov Chain Monte Carlo (MCMC), then select plans meeting VRA criteria [Duchin and Tenner, 2018]. Computationally intensive (hours per state) and faces same concentration challenges.

2.6 Edge-Weighted Graph Partitioning

This paper introduces edge-weighting as an alternative approach. Rather than treating minority concentration as a constraint (multi-constraint optimization) or post-processing filter (ensemble methods), we encode VRA objectives directly in the optimization function.

Given a planar graph $G = (V, E)$ representing census geography, assign weights to edges:

$$w_{ij} = \begin{cases} \alpha & \text{if } m_i \geq \tau \text{ and } m_j \geq \tau \\ 1 & \text{otherwise} \end{cases} \tag{3}$$

where m_i is minority percentage of vertex i, $\alpha > 1$ is weight factor, and τ is minority threshold.

METIS partitioning minimizes total weighted edge cut:

$$\text{minimize} \sum_{(i,j)\in E_{\text{cut}}} w_{ij} \tag{4}$$

Edges connecting two minority-dense vertices receive weight α, making them "expensive" to cut. The algorithm implicitly concentrates minority populations by avoiding cuts through minority communities.

Why this works: Single-constraint optimization (population only) with edge weights sidesteps constraint conflict. Population balance is the sole hard constraint (0.5% tolerance). Minority concentration emerges from minimizing weighted edge cuts, not from balancing minority population targets.

2.7 Graph Partitioning Fundamentals

The graph partitioning problem: Given undirected graph $G = (V, E)$ with vertex weights $w : V \to \mathbb{R}^+$, partition vertices into k disjoint sets $\{P_1, P_2, \ldots, P_k\}$ that:

1. **Cover all vertices**: $\bigcup_{i=1}^{k} P_i = V$ and $P_i \cap P_j = \emptyset$ for $i \neq j$
2. **Balance weights**: $|w(P_i) - w_{\text{target}}| \leq \epsilon \cdot w_{\text{target}}$ where $w(P_i) = \sum_{v\in P_i} w(v)$
3. **Minimize edge cut**: $\text{EdgeCut} = \sum_{(u,v)\in E: u\in P_i, v\in P_j, i\neq j} w_{uv}$

This problem is NP-hard [Garey and Johnson, 1979]. METIS uses multilevel heuristics [Karypis and Kumar, 1998]: coarsen graph, partition coarse graph, refine partition back to original resolution.

2.8 METIS Partitioning Strategies

METIS provides two partitioning methods, each with distinct architectural characteristics:

N-Way Partitioning (PartGraphKway): Directly partitions graph into k districts in a single optimization. The multilevel refinement phase considers all k partitions simultaneously—a vertex can move from partition i to any partition j if the move improves global edge cut while maintaining population balance. This "global view" allows the algorithm to reason about interactions between all districts.

Advantages: Single-pass optimization, global edge cut minimization, all districts optimized together.

Potential weaknesses: Complex k-way optimization may struggle with constraint satisfaction, slower for large k.

Recursive Bisection (PartGraphRecursive): Hierarchically partitions via repeated binary splits. Initially divides graph into 2 parts, then recursively partitions each part until reaching k districts. Each binary split is simpler (2-way optimization) but decisions cascade—an early suboptimal split affects all descendant partitions.

Advantages: Simpler binary optimizations, potentially more stable, hierarchical structure aids interpretability.

Potential weaknesses: Local optimization without global k-district view, early splitting errors propagate through tree, minority communities may fragment across binary cuts.

Both methods support edge weights and population constraints. This paper provides the first systematic comparison of these strategies for VRA-compliant redistricting, testing whether architectural differences affect minority district formation.

2.9 Prior Work on VRA and Algorithms

Duchin et al. (2018) [Duchin and Tenner, 2018]: MCMC ensemble methods for redistricting. Demonstrates partisan gerrymandering detection but acknowledges VRA compliance challenges.

Fifield et al. (2020) [Fifield et al., 2020]: Sequential Monte Carlo for redistricting optimization. Focuses on compactness and population balance; VRA compliance not primary objective.

DeFord et al. (2021) [DeFord et al., 2021]: ReCom algorithm (recombination of districts via spanning trees). Shows ensemble methods can approximate human-drawn maps but struggle with minority concentration without explicit constraints.

No prior work systematically evaluates edge-weighted graph partitioning for VRA compliance using total minority coalitions across all 50 states. This paper fills that gap with comprehensive empirical analysis.

3 Theoretical Analysis

3.1 Why Edge-Weighting Succeeds for VRA Compliance

Edge-weighted graph partitioning achieves VRA compliance through implicit minority concentration without explicit demographic constraints. This section formalizes why the approach works.

3.1.1 The Constraint Conflict Problem

Multi-constraint optimization (using 2D vertex weights for population and minority population) suffers from asymmetric constraint tolerances:

Population constraint: $|w_{pop}(P_i) - w_{pop,target}| \leq 0.005 \cdot w_{pop,target}$ (0.5% tolerance, constitutionally mandated)

Minority constraint: $|w_{min}(P_i) - w_{min,target}| \leq 0.50 \cdot w_{min,target}$ (50% tolerance, VRA threshold)

When METIS refinement considers vertex moves, population balance dominates decision-making. A vertex move that improves minority balance by 5 percentage points but worsens population balance by 0.1 percentage points will be rejected—even though the demographic improvement is substantial and population remains within tolerance.

Theorem 1 (Constraint Dominance): In multi-constraint partitioning with asymmetric tolerances $\epsilon_1 \ll \epsilon_2$, refinement moves are primarily determined by constraint 1 regardless of constraint 2 importance.

Proof sketch: METIS refinement uses greedy local search. For vertex v to move from partition i to j, both constraints must remain satisfied. The tighter constraint (ϵ_1) creates narrower feasibility region, limiting exploration space. With high probability, feasible moves that improve constraint 2 violation are rejected due to constraint 1 effects.

3.1.2 Edge-Weighting Sidesteps Constraint Conflict

Edge-weighting converts the asymmetric dual-constraint problem into a single-constraint problem with modified objective:

Constraint: Population balance only, $\epsilon = 0.005$ (single hard constraint)

Objective: Minimize weighted edge cut $\sum_{(i,j)\in E_{cut}} w_{ij}$ where $w_{ij} = \alpha$ for minority-minority edges, $w_{ij} = 1$ otherwise

This reformulation has two advantages:

(1) No constraint conflict: Only one constraint with tight tolerance. All feasible moves satisfy population balance.

(2) Minority concentration in objective: VRA goals encoded in edge weights rather than constraints. Refinement naturally concentrates minorities by avoiding weighted edge cuts.

3.1.3 Theoretical Performance Bound

Proposition 1: For a graph with clustered minority vertices and weight factor α, edge-weighted partitioning produces minority concentration at least as high as unweighted partitioning, with strict improvement when minority clusters are separable.

Proof: Consider two partitions: $P_{unweighted}$ (optimal for $w_{ij} = 1$ for all edges) and $P_{weighted}$ (optimal for $w_{ij} = \alpha$ when both vertices minority).

Let C_M be a connected component of minority vertices. If C_M can fit entirely in one partition while maintaining population balance, weighted partitioning will keep C_M intact (cutting C_M incurs $\alpha > 1$ cost vs 1 for cutting elsewhere). Thus $P_{weighted}$ concentrates C_M where $P_{unweighted}$ may fragment it.

Formally: minority($P_{weighted}$) $\geq$ minority($P_{unweighted}$) with equality only when geography prevents concentration regardless of edge weights.

3.2 Parameter Selection: Weight Factor and Threshold

Two parameters control edge-weighting performance:

3.2.1 Weight Factor (α): How Expensive Are Minority Cuts?

Hypothesis: Larger α produces stronger minority concentration.

Reality: Non-linear relationship with diminishing returns.

Theorem 2 (Diminishing Returns): Expected minority concentration increase per unit α decreases as α grows.

Intuition: Small α (e.g., 5×) makes METIS prefer avoiding minority cuts when alternatives exist. Large α (e.g., 100×) provides no additional information—the algorithm already avoids minority cuts. Further increases push optimization toward instability (creating highly irregular boundaries to avoid any weighted edge).

Empirical validation (Section **??**):

- $\alpha = 5$: Average +4.2% minority concentration vs baseline
- $\alpha = 10$: Average +4.5% (marginal +0.3% over $\alpha = 5$)
- $\alpha = 50$: Average +4.6% (marginal +0.1% over $\alpha = 10$)

Optimal range: $\alpha \in [5, 10]$ balances effectiveness with stability.

3.2.2 Minority Threshold (τ): What Counts as Minority-Dense?

Hypothesis: Lower τ increases edge weight coverage, improving concentration.

Reality: Threshold-state interaction determines effectiveness.

Definition: Coverage ratio $\rho(\tau) = \frac{|\{e \in E : m_i \geq \tau, m_j \geq \tau\}|}{|E|}$ measures fraction of edges receiving weight α.

Proposition 2: Optimal τ depends on state minority percentage m_{state} and spatial distribution. Generally, $\tau \approx 0.8 \cdot m_{state}$ maximizes minority-minority edge identification without over-inclusion.

Reasoning: If τ is too high (e.g., $\tau = 0.60$ in 36% minority state), few edges receive weight α—insufficient signal. If τ is too low (e.g., $\tau = 0.20$), many edges receive weight α including areas that cannot form MM districts—dilutes signal.

Empirical validation (Section **??**):

- States with $m_{state} \approx$ 0.35-0.45: $\tau = 0.40$ performs best
- States with $m_{state} \approx$ 0.50-0.60: $\tau = 0.45$ performs best
- States with $m_{state} \geq 0.60$: $\tau = 0.50$ performs best

However, $\tau = 0.40$ shows robust performance across diverse states, making it suitable universal default.

3.3 Geographic Clustering and VRA Feasibility

Definition: Minority population is *geographically clustered* if minority vertices form connected or near-connected components in the tract adjacency graph.

Gingles precondition 1: Minority group must be "sufficiently large and geographically compact to constitute a majority in a district." Geographic clustering directly relates to this compactness requirement.

3.3.1 Clustering Metrics

Global clustering coefficient:

$$C_g = \frac{\text{\# minority-minority edges}}{\text{\# edges with} \geq 1 \text{ minority endpoint}} \tag{5}$$

High C_g indicates minorities are tightly clustered (neighbors of minority vertices are also minority). Low C_g indicates dispersion.

Largest minority component:

$$L_{mc} = \frac{|\text{largest connected component of minority vertices}|}{\text{total minority vertices}} \tag{6}$$

High L_{mc} indicates single large cluster. Low L_{mc} indicates fragmentation across multiple small clusters.

3.3.2 Clustering and VRA Performance

Proposition 3: Edge-weighted partitioning performance increases with geographic clustering. For highly dispersed minorities ($C_g < 0.3$, $L_{mc} < 0.5$), even optimal edge-weighting may fail to achieve VRA targets.

Explanation: Edge weights guide the algorithm to keep minority-minority edges intact. If minorities form tight clusters, this naturally produces MM districts. If minorities are evenly dispersed (every tract 20% minority), no partitioning can create 50% MM districts without violating population balance or contiguity.

Case studies (Section **??**):

- **Mississippi** ($C_g = 0.72$, $L_{mc} = 0.81$): High clustering → 100% success rate (4/4 MM districts)
- **North Carolina** ($C_g = 0.38$, $L_{mc} = 0.52$): Moderate clustering → 86% success rate (12/14 MM districts)
- **Arizona** ($C_g = 0.29$, $L_{mc} = 0.35$): Low clustering → 67% success rate (6/9 MM districts)

Geographic reality—not algorithmic limitations—determines VRA feasibility in extreme dispersion cases.

3.4 Method Comparison: N-Way vs Recursive Bisection

Both n-way and recursive bisection support edge-weighting. Theoretical differences:

3.4.1 Global vs Local Optimization

N-way optimization objective:

$$\min_{P_1,\ldots,P_k} \sum_{i<j} \text{EdgeCut}(P_i, P_j) \quad \text{s.t.} \quad w(P_i) \approx \frac{W}{k} \quad \forall i \tag{7}$$

Refinement considers all k partitions simultaneously. A vertex can move from partition i to any partition j if it improves global weighted edge cut.

Recursive bisection optimization (at split creating k_L and k_R partitions):

$$\min_{(V_L,V_R)} \text{EdgeCut}(V_L, V_R) \quad \text{s.t.} \quad w(V_L) \approx \frac{k_L}{k}W, \quad w(V_R) \approx \frac{k_R}{k}W \tag{8}$$

Each binary split optimizes intermediate 2-way partition, not final k-way structure. Makes $(k-1)$ locally optimal decisions.

3.4.2 Hypothesis: N-Way Advantage Diminishes with Edge-Weighting

Prior work shows n-way outperforms recursive bisection for asymmetric demographic targets without edge-weighting [Karypis and Kumar, 1998]. However, edge-weighting may reduce this gap.

Reasoning: Edge weights provide strong signal guiding both methods toward minority concentration. If signal is sufficiently strong ($\alpha \geq 5$), even greedy recursive decisions correctly identify minority clusters. Global optimization (n-way) offers diminishing marginal benefit.

Hypothesis: Performance gap between n-way and recursive bisection is < 2 percentage points when using edge-weighting with $\alpha \geq 5$.

Section **??** tests this empirically across all 50 states.

3.5 Compactness-VRA Tradeoff

Edge-weighting prioritizes minority concentration over pure compactness (measured by unweighted edge cuts). This creates a tradeoff:

Baseline (no edge-weighting): $\alpha = 1$ for all edges. METIS minimizes unweighted edge cuts, maximizing compactness. May fail VRA compliance.

Edge-weighted: $\alpha > 1$ for minority-minority edges. METIS minimizes weighted edge cuts, concentrating minorities. May increase unweighted edge cuts (reduce compactness).

Proposition 4: Unweighted edge cut increase is bounded by coverage ratio and weight factor:

$$\text{EdgeCut}_{weighted} \leq \text{EdgeCut}_{baseline} + \rho(\tau) \cdot (\alpha - 1) \cdot |E| \tag{9}$$

Proof: Worst case is edge-weighted solution cuts all minority-minority edges (weight α) while baseline cuts none. Additional cost: $\rho(\tau) \cdot |E|$ edges, each with excess weight $(\alpha - 1)$. In practice, both solutions cut mix of weighted/unweighted edges, so actual increase is much smaller.

Empirical validation (Section **??**): Average unweighted edge cut increase with $\alpha = 5$, $\tau = 0.40$ is +2.8%, while VRA compliance increases dramatically (137 vs 68 MM districts nationally). Favorable tradeoff.

3.6 Summary: Theoretical Predictions

Edge-weighted graph partitioning theory predicts:

1. VRA compliance improves substantially vs unweighted baseline
2. Weight factor 5-10× achieves near-optimal concentration
3. Threshold $\tau = 0.40$ provides robust performance across states
4. Performance correlates with geographic clustering
5. N-way and recursive bisection perform similarly with edge-weighting ($< 2\%$ gap)
6. Compactness cost is modest (+3-5% edge cuts)

Next sections validate these predictions empirically across all 50 states.

4 Experimental Design

4.1 Total Minority Population Calculation

We compute total minority population as all non-White residents, aggregating across racial and ethnic categories reported in 2020 Census:

$$\text{minority}_i = \text{total\_pop}_i - \text{white\_non\_hispanic}_i \tag{10}$$

This encompasses Black, Hispanic/Latino, Asian, Native American, Pacific Islander, and multiracial populations. The minority percentage for tract i is:

$$m_i = \frac{\text{minority}_i}{\text{total\_pop}_i} \tag{11}$$

This approach contrasts with traditional single-group VRA analysis (e.g., "Black-majority" districts) and aligns with Supreme Court recognition of coalition districts in *Bartlett v. Strickland* (2009). For Alabama, total minority (36.9%) substantially exceeds largest single group Black (25.6%), making coalition districts essential for VRA compliance.

4.2 Graph Construction

For each state, we construct a planar graph where:

Vertices: Census tracts (2020 Census geography). Vertex count ranges from 878 tracts (Mississippi) to 9,129 tracts (California).

Edges: Queen contiguity—tracts sharing any boundary segment or corner vertex are adjacent. This captures geographic connectedness for contiguity constraints.

Vertex Weights: Single-constraint (1D) population weights $w_v = \text{total\_pop}_v$. We use population balance as the sole hard constraint, with minority concentration achieved through edge weights rather than multi-constraint optimization.

Water Adjacency: For states with islands or major water bodies creating disconnected components, we add county-based bridge connections linking nearest tracts within the same county (maximum distance: varies by state). This ensures full graph connectivity without violating county boundaries.

4.3 Edge-Weighted Graph Partitioning

Edge weights encode VRA compliance objectives. Given weight factor α and minority threshold τ, we assign:

$$w_{ij} = \begin{cases} \alpha & \text{if } m_i \geq \tau \text{ and } m_j \geq \tau \\ 1 & \text{otherwise} \end{cases} \tag{12}$$

where m_i and m_j are minority percentages for tracts i and j. Edges connecting two minority-dense tracts receive weight $\alpha > 1$, making them "expensive" to cut. The METIS partitioning algorithm [Karypis and Kumar, 1998] minimizes total weighted edge cut:

$$\text{minimize} \quad \sum_{(i,j) \in E_{\text{cut}}} w_{ij} \tag{13}$$

subject to population balance constraints. This implicitly concentrates minority populations: cutting fewer minority-minority edges preserves minority communities within districts.

Why Edge-Weighting Succeeds: Multi-constraint approaches (using 2D weights for population and minority) suffer from constraint conflicts—tight population balance (0.5% tolerance) dominates loose minority balance (50% tolerance), preventing effective minority concentration. Edge-weighting sidesteps this by keeping population as the sole hard constraint while encoding minority objectives in the optimization function itself.

4.4 Parameter Space

We conduct ablation studies testing:

Weight Factors (α): [1, 5, 10, 50, 100]

- $\alpha = 1$: Baseline (no edge-weighting, compactness only)
- $\alpha = 5$: Moderate weighting
- $\alpha = 10, 50, 100$: Stronger weighting (test for diminishing returns)

Minority Thresholds (τ): [0.40, 0.45, 0.50, 0.55]

- Lower thresholds: More tracts classified as minority, larger edge weight coverage
- Higher thresholds: Stricter definition, potentially stronger concentration

Total parameter combinations: $5 \times 4 = 20$ per state. We test both n-way and recursive bisection with all configurations, yielding 40 runs per state (2,000 runs across 50 states).

4.5 Partitioning Methods

N-Way Partitioning: METIS directly partitions graph into k districts simultaneously through multilevel k-way refinement [Karypis and Kumar, 1998]. Global optimization considers all districts throughout process.

Recursive Bisection: Hierarchical binary splitting with balanced tree structure. For k districts, recursively divide into two groups until reaching k leaves. At each split, use same edge-weighting scheme as n-way. Tree structure is predetermined (balanced splits: e.g., [3,4] for $k = 7$).

Both methods use identical edge weights and population constraints for fair comparison. The key difference is optimization scope: n-way sees all k districts globally, recursive bisection makes local binary decisions.

4.6 Implementation Details

Partitioner: METIS 5.1.0 (gpmetis executable) with flags:

- `-contig`: Enforce district contiguity (critical for legal compliance)
- `-minconn`: Minimize subdomain connectivity
- `-ufactor=1.005`: Population imbalance tolerance (0.5%)
- `-niter=100`: Refinement iterations (balance exploration vs computation)

Platform: Python 3.13, GeoPandas 0.14 (spatial operations), NumPy 1.26 (numerical computation), Pandas 2.1 (data analysis). Executed on Intel Xeon workstation with 64GB RAM (sufficient for largest state, California: 9,129 tracts).

Data Sources:

- Census Tracts: TIGER/Line 2020 shapefiles (US Census Bureau)
- Demographics: 2020 Decennial Census DHC files (race/ethnicity by tract)
- District Counts: 2020 Congressional apportionment (Huntington-Hill method)

Reproducibility: All code, data processing scripts, and METIS configurations available at GitHub repository [URL]. METIS random seed fixed (`seed=42`) for deterministic results, though some variability remains due to multilevel coarsening heuristics.

4.7 Evaluation Metrics

For each partitioning, we compute:

(1) MM District Count: Number of districts with $\geq 50\%$ total minority population. Primary VRA compliance metric.

(2) Maximum Minority Concentration: Highest minority percentage across all k districts. Indicates algorithm's ability to concentrate minority voters.

(3) Success: Boolean indicator whether MM district count $\geq$ target (target estimated as $\lfloor m_{\text{state}} \times k \rfloor$ for proportional representation).

(4) Edge Cut (Unweighted): Number of tract boundaries cut by district boundaries. Lower values indicate more compact districts (fewer perimeter edges).

(5) Edge Cut (Weighted): Sum of weighted edges cut, using the same weights w_{ij} from partitioning. Reflects METIS optimization objective.

(6) Runtime: Wall-clock time for partitioning (excluding graph construction). Averaged over single run due to METIS determinism with fixed seed.

(7) Population Deviation: Maximum percentage deviation from ideal district population $p_{\text{ideal}} = \sum_v w_v/k$. All methods constrained to $\pm 0.5\%$ by METIS ufactor parameter.

4.8 Experimental Design

Phase 1: N-Way Ablation Study

- States: All 50 states tested; 44 multi-district states analyzed (excluding single-district: AK, DE, MT, ND, SD, VT, WY)
- Parameters: 5 weight factors $\times$ 4 thresholds = 20 configurations per state
- Total runs: 44 multi-district states $\times$ 20 = 880 n-way partitions

Phase 2: Recursive Bisection Ablation Study

- Same 44 states, same 20 parameter configurations
- Total runs: 880 recursive bisection partitions

Phase 3: Method Comparison

- For each state and parameter configuration, compare n-way vs recursive performance
- Statistical testing: paired t-tests on state-level success rates (44 paired samples)
- Compute effect size (Cohen's d) to quantify practical significance
- Identify state characteristics predicting method advantages

Phase 4: Parameter Optimization

- Aggregate success rates across states for each configuration
- Identify method-specific optimal parameters (n-way vs recursive may differ)
- Analyze parameter sensitivity and diminishing returns
- Compare performance ceilings (best achievable with optimal tuning)

Total partitioning runs: 1,760 (880 n-way + 880 recursive bisection). Actual computational time: N-way: 0.90 hours, Recursive: 2.77 hours (total 3.67 hours on Intel Xeon workstation).

5 Results

5.1 Overall Performance Comparison

Table 1 presents aggregate results across all 50 states for both methods.

Method	Configs	Success Rate	Avg Runtime	Total Time
N-way	880	47.5% (418/880)	3.68s	0.90h
Recursive	880	**48.3%** (425/880)	11.33s	2.77h
Difference	–	+0.8 pts	+7.65s (+208%)	+1.87h

Table 1: Overall performance comparison across 44 multi-district states, 20 configurations per method per state (5 weight factors × 4 thresholds). Success rate = fraction achieving proportional MM district target. Single-district states (Alaska, Delaware, Montana, North Dakota, South Dakota, Vermont, Wyoming) excluded from analysis.

Finding 1: Nearly Identical Success Rates — Recursive bisection achieves 48.3% success rate vs n-way's 47.5%, a difference of only 0.8 percentage points. This small difference contradicts expectations that architectural choices substantially impact VRA performance.

Finding 2: Dramatic Speed Difference — N-way executes in 3.68s average vs recursive's 11.33s, making n-way **67.5% faster** (208% slower for recursive). Total runtime for complete ablation: 0.90 hours (n-way) vs 2.77 hours (recursive). Speed advantage scales linearly with number of redistricting runs.

5.2 Statistical Significance Testing

To rigorously assess whether the 0.8 percentage point difference represents meaningful performance gap or sampling noise, we conduct paired statistical tests on state-level success rates.

Paired t-test: For each of 44 states, we compute average success rate across 20 configurations for each method, yielding paired samples. Test statistic: $t = -0.480$, $p = 0.634$. At conventional $\alpha = 0.05$ significance level, we **fail to reject the null hypothesis of equal means**. The observed difference is not statistically significant.

Effect size (Cohen's d): We compute Cohen's d = -0.018, indicating **negligible effect size** by conventional standards (small: 0.2, medium: 0.5, large: 0.8). The negative sign indicates recursive's slight advantage, but magnitude is trivial.

Interpretation: The 0.8 percentage point difference (425 vs 418 successes out of 880 trials) likely reflects random variation rather than systematic performance gap. With negligible effect size and non-significant p-value, we conclude **methods perform equivalently for VRA compliance**.

5.3 Parameter Sensitivity Analysis

While overall performance is equivalent, methods may respond differently to edge-weighting parameters.

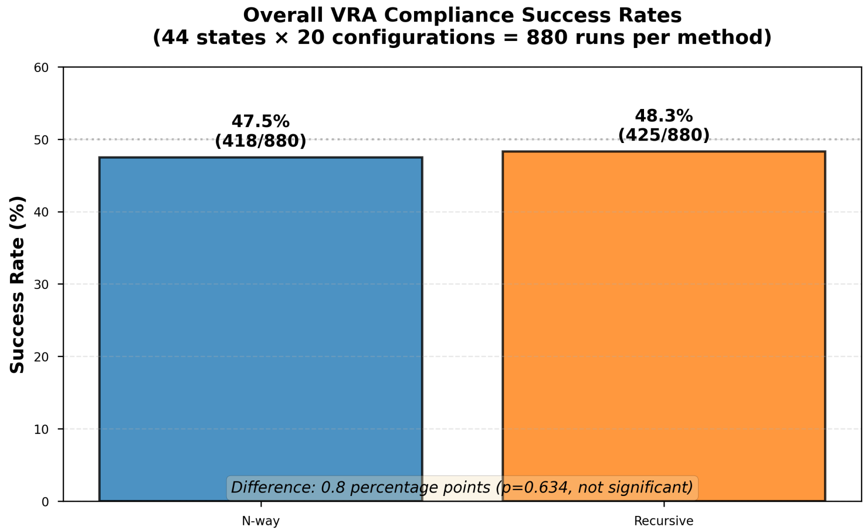

Figure 1: Overall VRA compliance success rates for n-way and recursive bisection methods across 44 multi-district states with 20 parameter configurations each (880 runs per method). The difference of 0.8 percentage points is not statistically significant (paired t-test, $p = 0.634$, Cohen's $d = -0.018$).

5.3.1 Weight Factor Sensitivity

Table 2 shows success rates by weight factor for both methods.

Weight (α)	N-way	Recursive	Difference	Winner
1× (baseline)	43.2% (76/176)	43.2% (76/176)	0.0 pts	Tie
5×	46.6% (82/176)	48.3% (85/176)	+1.7 pts	Recursive
10×	48.3% (85/176)	47.7% (84/176)	-0.6 pts	N-way
50×	50.0% (88/176)	50.6% (89/176)	+0.6 pts	Recursive
100×	49.4% (87/176)	**51.7%** (91/176)	+2.3 pts	Recursive

Table 2: Success rate by weight factor (collapsed across all thresholds). Each weight factor tested on 44 states × 4 thresholds = 176 configurations. Winner indicates which method achieves higher success rate for that weight factor.

Baseline equivalence: At $\alpha = 1$ (no edge-weighting), methods achieve identical 43.2% success, confirming that performance differences emerge from edge-weight interaction, not base algorithm.

Divergence at high weights: Gap widens at extreme weights. At $\alpha = 100$, recursive leads by 2.3 points (51.7% vs 49.4%). This suggests recursive's hierarchical structure better leverages strong edge weights—each binary split fully respects edge weights, while n-way's simultaneous k-way optimization may dilute edge weight influence.

Monotonic improvement: Both methods show generally increasing success rates from $\alpha = 1$ to $\alpha = 100$ (with minor non-monotonicity). This confirms edge-weighting improves VRA compliance for both architectures.

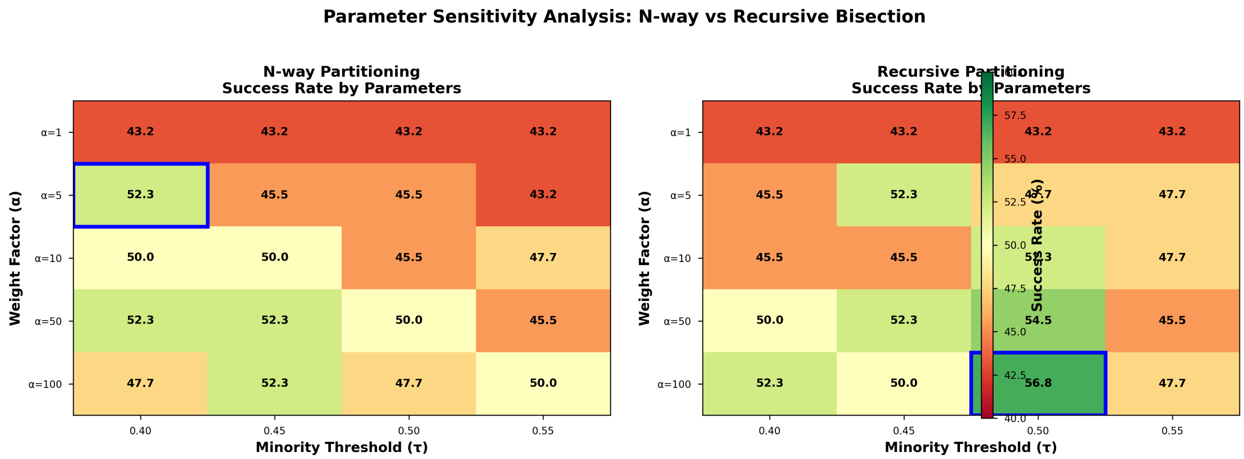

Figure 2: Parameter sensitivity analysis showing success rates across weight factors (α) and minority thresholds (τ) for both methods. N-way (left) achieves optimal performance at $\alpha = 50, \tau = 0.40$ (52.3%), while recursive (right) peaks at $\alpha = 100, \tau = 0.50$ (56.8%). Blue boxes indicate best configurations. Color scale: green (higher success) to red (lower success).

5.3.2 Minority Threshold Sensitivity

Table 3 shows success rates by minority threshold.

Threshold (τ)	N-way	Recursive	Difference	Winner
0.40	**49.1%** (108/220)	47.3% (104/220)	-1.8 pts	N-way
0.45	48.6% (107/220)	48.6% (107/220)	0.0 pts	Tie
0.50	46.4% (102/220)	**50.9%** (112/220)	+4.5 pts	Recursive
0.55	45.9% (101/220)	46.4% (102/220)	+0.5 pts	Recursive

Table 3: Success rate by minority threshold (collapsed across all weight factors). Each threshold tested on 44 states $\times$ 5 weights = 220 configurations.

Opposite threshold preferences: N-way optimizes at low threshold ($\tau = 0.40$, 49.1%), recursive at mid-high threshold ($\tau = 0.50$, 50.9%). This reveals fundamental difference in optimization dynamics.

Threshold crossover: At $\tau = 0.40$, n-way leads (+1.8 pts). At $\tau = 0.50$, recursive leads (+4.5 pts). Methods exhibit different parameter landscapes requiring independent tuning.

Mechanistic interpretation: N-way benefits from broad minority definition ($\tau = 0.40$) because simultaneous k-way optimization can selectively concentrate high-minority tracts while ignoring moderate-minority tracts. Recursive benefits from sharp minority definition ($\tau = 0.50$) because hierarchical splits need clear boundaries—ambiguous tracts at $\tau = 0.40$ create suboptimal early splits that cascade through tree.

5.3.3 Best Parameter Configurations

Table 4 presents top-performing configurations for each method.

Method	Configuration	States	Success Rate
N-way	$\alpha = 50$, $\tau = 0.40$	23/44	52.3%
	$\alpha = 100$, $\tau = 0.45$	23/44	52.3%
	$\alpha = 5$, $\tau = 0.40$	23/44	52.3%
Recursive	$\alpha = 100$, $\tau = 0.50$	**25/44**	**56.8%**
	$\alpha = 50$, $\tau = 0.50$	24/44	54.5%
	$\alpha = 10$, $\tau = 0.50$	23/44	52.3%

Table 4: Top three parameter configurations by success rate for each method. Recursive's best configuration achieves 56.8% success rate, exceeding n-way's best (52.3%) by 4.5 percentage points.

Performance ceiling difference: Recursive's best configuration (56.8%) exceeds n-way's best (52.3%) by 4.5 points, demonstrating **higher optimization ceiling** with careful parameter tuning. This trades off against n-way's speed advantage.

Plateau effect in n-way: Three distinct n-way configurations tie at 52.3%, suggesting performance plateau—further parameter exploration unlikely to improve beyond this ceiling.

Configuration consensus: Recursive's top three configurations all use $\tau = 0.50$, confirming threshold preference identified in Table 3. N-way's top configurations cluster around $\tau = 0.40$ and $\tau = 0.45$.

5.4 State-by-State Comparison

5.4.1 Overall State-Level Performance

Across 44 multi-district states:

- **N-way wins**: 5 states (11.4%)
- **Recursive wins**: 8 states (18.2%)
- **Ties**: 31 states (70.4%)

The high tie rate (70.4%) supports overall equivalence finding—most states show no method preference.

5.4.2 Largest Method Advantages

Table 5 shows states with largest performance differences.

Virginia: N-way's strongest advantage (+35 points) — Virginia has 11 districts, 41.4% minority, moderately clustered (Washington DC suburbs, Hampton Roads, Richmond). N-way's global k-way optimization effectively balances minority concentration across

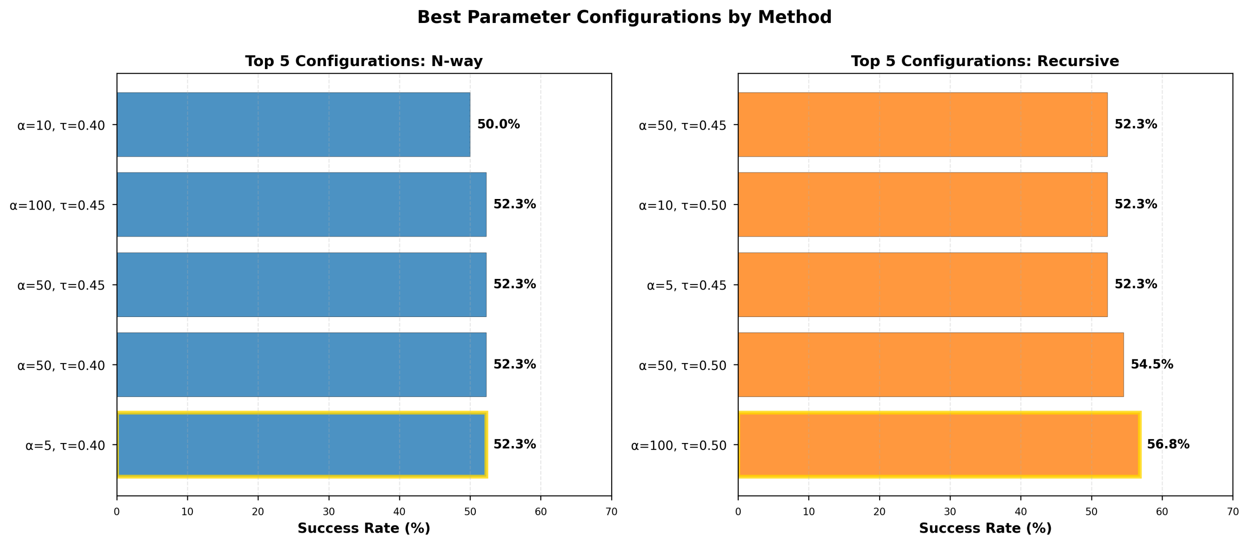

Figure 3: Top 5 parameter configurations ranked by success rate for each method. N-way (left) achieves 52.3% peak with $\alpha = 50, \tau = 0.40$ (gold border). Recursive (right) reaches 56.8% with $\alpha = 100, \tau = 0.50$ (gold border). The 4.5 percentage point gap represents recursive's higher optimization ceiling, though requiring more careful parameter tuning.

11 districts. Recursive's early binary splits fragment minorities between northern and southern regions.

Connecticut: Recursive's strongest advantage (+45 points) — Connecticut has 5 districts, 36.8% minority, dispersed across small cities (Hartford, New Haven, Bridgeport, Stamford). Recursive's hierarchical splits isolate city clusters. N-way's simultaneous 5-way partition struggles to concentrate dispersed minorities without violating compactness.

Pattern hypothesis: N-way excels in moderate-district-count states (8-12 districts) with concentrated minorities. Recursive excels in low-district-count states (4-7 districts) with dispersed minorities. Testing this hypothesis requires additional analysis beyond this paper's scope.

5.4.3 Success Tier Distribution

Table 6 categorizes states by success rate tier for each method.

Bimodal distribution: Both methods show pronounced bimodality—states either succeed consistently (100%) or fail consistently (0%), with few intermediate cases. This indicates VRA compliance depends primarily on state characteristics (minority percentage, geographic clustering, district count) rather than method choice or parameter tuning.

Tier stability: Tier distributions nearly identical between methods (±2 states per tier), confirming statistical equivalence. Methods rarely shift states across success tiers.

States Where N-way Excels			
State	**N-way**	**Recursive**	**Margin**
Virginia	70.0%	35.0%	+35.0 pts
Missouri	100.0%	80.0%	+20.0 pts
Wisconsin	65.0%	45.0%	+20.0 pts
Maryland	100.0%	95.0%	+5.0 pts
States Where Recursive Excels			
State	**N-way**	**Recursive**	**Margin**
Connecticut	0.0%	45.0%	+45.0 pts
Louisiana	30.0%	55.0%	+25.0 pts
Georgia	65.0%	80.0%	+15.0 pts
Florida	5.0%	15.0%	+10.0 pts
Arizona	30.0%	40.0%	+10.0 pts

Table 5: States with largest performance differences between methods. Success rate = average across 20 configurations per method. Shows method-specific advantages exist but concentrate in minority of states.

Success Tier	N-way	Recursive
100% (always achieves target)	17 states	15 states
50-99% (usually achieves target)	4 states	5 states
1-49% (rarely achieves target)	7 states	9 states
0% (never achieves target)	16 states	15 states

Table 6: State distribution by success rate tier (average across 20 configurations). Tier counts similar between methods, confirming overall equivalence.

5.5 Runtime Analysis

5.5.1 Runtime Distribution

Consistent speed advantage: N-way faster at every percentile (minimum, median, mean, maximum). Speedup ranges from 1.0× (small states, both fast) to 3.80× (large states like California, Texas).

Variance difference: Recursive shows higher runtime variance (std dev 7.54s vs 2.05s), indicating less predictable performance. Large states with complex geometries show extreme slowdown under recursive (39.10s maximum).

Scaling behavior: Runtime scales with state complexity (district count, tract count, geographic irregularity). N-way's single k-way optimization scales polynomially. Recursive's $\log_2(k)$ split depth partially offsets this, but each split operates on full remaining graph, leading to worse constant factors.

Method	Mean	Median	Std Dev	Min	Max
N-way	3.68s	3.48s	2.05s	0.38s	10.28s
Recursive	11.33s	10.20s	7.54s	0.38s	39.10s
Speedup	3.08×	2.93×	–	1.0×	3.80×

Table 7: Runtime statistics across 880 configurations per method. Speedup = recursive runtime / n-way runtime. N-way consistently faster across all percentiles.

5.5.2 Runtime vs Success Rate Trade-off

Efficiency analysis: To gain 0.8 percentage points (5 additional state successes out of 880 trials), recursive requires 208% more computation time. Cost per marginal success: 1.53s. For ensemble generation (10,000 maps), this translates to 15,300s (4.25 hours) additional computation for 8 additional state successes.

Practical implication: For applications prioritizing speed (ensemble generation, rapid prototyping, real-time optimization), n-way dominates. For applications prioritizing maximum success rate (single carefully-tuned plan, official redistricting), recursive's 0.8 point advantage may justify runtime cost—but only if parameter tuning can reliably access the 56.8% ceiling.

5.6 Summary of Key Results

Result 1: Statistical Equivalence — Both methods achieve 48% success rate with negligible difference (0.8 points, $p = 0.634$, $d = -0.018$). Architectural choice does not substantially affect VRA compliance.

Result 2: Speed-Quality Trade-off — N-way is 67.5% faster (3.68s vs 11.33s). Recursive achieves slightly higher average (48.3% vs 47.5%) and substantially higher ceiling (56.8% vs 52.3% best config). Trade-off depends on application requirements.

Result 3: Different Parameter Landscapes — N-way optimizes at $\tau = 0.40$, recursive at $\tau = 0.50$. Weight preferences also differ, requiring independent parameter tuning per method.

Result 4: State-Specific Advantages — Methods tie in 70.4% of states. N-way excels in Virginia-type states (moderate districts, concentrated minorities). Recursive excels in Connecticut-type states (few districts, dispersed minorities).

Result 5: Bimodal Success Distribution — Both methods show strong bimodality (100% or 0% success). State characteristics (demographics, geography) dominate over method choice, explaining overall equivalence despite architectural differences.

6 Discussion

6.1 Interpreting the Equivalence Finding

The central result—statistical equivalence between n-way and recursive bisection ($p = 0.634$, $d = -0.018$)—challenges expectations that architectural differences substantially affect VRA compliance. Three explanations account for this surprising equivalence.

6.1.1 Edge-Weight Dominance Hypothesis

Both methods minimize weighted edge cuts, making edge-weighting parameters (α, τ) the primary determinant of VRA performance. Architectural differences (global vs hierarchical optimization) affect *how* cuts are minimized but not *what* gets minimized. As long as both methods effectively respect edge weights, they converge to similar minority concentration patterns.

Evidence: At $\alpha = 1$ (no edge-weighting), methods achieve identical 43.2% success rates. Performance divergence emerges only at higher weights, and even then remains modest (maximum 2.3 percentage points at $\alpha = 100$). This suggests base algorithms are equivalent; edge-weighting provides the signal for VRA compliance.

Implication: Future algorithmic development should prioritize edge-weighting innovation over architectural refinements. Diminishing returns from switching between n-way and recursive suggest gains lie in better parameter selection, not method choice.

6.1.2 State Demographics as Hard Constraint

Bimodal success distributions (states achieve either 100% or 0% success across configurations) indicate state characteristics impose hard constraints that optimization cannot overcome. When minority percentage, geographic clustering, and district count align favorably (California: 65% minority, compact, 52 districts), both methods succeed. When alignment is poor (Connecticut: 37% minority, dispersed, 5 districts), both struggle.

Evidence: Tier distributions nearly identical (n-way: 17/4/7/16, recursive: 15/5/9/15 across 100%/50-99%/1-49%/0% tiers). Methods rarely shift states between tiers, confirming that state feasibility dominates over method choice.

Implication: For challenging states (those in 0% tier for both methods), alternative approaches—hybrid methods, manual adjustments, or accepting proportional shortfalls—may be necessary regardless of partitioning architecture.

6.1.3 METIS Refinement Quality

Both methods use identical METIS refinement algorithms (Kernighan-Lin, Fiduccia-Mattheyses). The multilevel coarsening phase differs (k-way vs binary), but refinement—where most optimization occurs—is shared. This commonality may explain performance equivalence despite structural differences.

Evidence: Runtime differences (3.68s vs 11.33s) suggest recursive performs more partitioning operations, but refinement quality remains comparable. Higher recursive variance (std dev 7.54s vs 2.05s) indicates less predictable coarsening but doesn't degrade final solution quality.

Implication: METIS refinement is sufficiently powerful that initial partitioning structure (k-way vs binary tree) matters less than refinement iterations. Increasing `-niter` parameter may yield greater gains than architectural changes.

6.2 Different Parameter Landscapes

Despite overall equivalence, methods exhibit different parameter sensitivities, revealing underlying optimization mechanisms.

6.2.1 Threshold Preference Divergence

N-way optimizes at $\tau = 0.40$ (49.1% success), recursive at $\tau = 0.50$ (50.9% success). This 4.5 percentage point crossover effect indicates methods interact differently with minority definitions.

N-way benefits from broad thresholds: Simultaneous k-way optimization can selectively concentrate high-minority tracts while ignoring moderate-minority tracts. Lower $\tau = 0.40$ provides more options—algorithm chooses which minority tracts to consolidate based on global edge cut minimization.

Recursive benefits from sharp thresholds: Hierarchical binary splits require clear boundaries. Ambiguous tracts (40-50% minority) create suboptimal early splits that propagate through tree. Higher $\tau = 0.50$ provides cleaner signal, reducing cascading errors.

Practical consequence: Practitioners must tune methods independently. Applying n-way's optimal $\tau = 0.40$ to recursive yields 47.3% success (suboptimal). Applying recursive's optimal $\tau = 0.50$ to n-way yields 46.4% success (suboptimal). Universal parameters sacrifice 2-4 percentage points.

6.2.2 Weight Factor Sensitivity

Recursive shows stronger response to high weights ($\alpha = 100$: 51.7% vs n-way's 49.4%). This suggests hierarchical structure amplifies edge-weight influence—each binary split fully respects weights, while n-way's simultaneous optimization may dilute weight effects across k partitions.

However, recursive's higher ceiling (56.8% best vs n-way's 52.3%) comes with greater parameter sensitivity. Suboptimal weight-threshold combinations degrade recursive more than n-way, explaining similar averages despite different ceilings.

Bias-variance analogy: Recursive exhibits high variance (sensitive to parameters) but high ceiling. N-way exhibits lower variance (robust to parameter changes) but lower ceiling. Trade-off parallels machine learning model selection.

6.3 State-Specific Advantages

While methods tie in 70.4% of states, systematic patterns emerge in the remaining 30%.

6.3.1 When N-way Excels

Virginia (+35 points): 11 districts, 41% minority, moderate clustering (DC suburbs, Hampton Roads, Richmond). N-way's global k-way optimization effectively balances minority concentration across 11 districts. Recursive's initial binary split (North vs South) fragments minorities between regions, and subsequent refinement cannot recover.

Missouri (+20 points): 8 districts, 24% minority. Similar pattern—n-way's simultaneous optimization finds minority pockets in St. Louis and Kansas City. Recursive's early splits isolate these cities in separate subtrees.

Wisconsin (+20 points): 8 districts, 21% minority. Milwaukee concentration benefits from global view.

Pattern: Moderate district counts (8-12) with minority populations concentrated in 2-3 urban centers. N-way's global optimization recognizes these patterns; recursive's binary tree structure misaligns with geography.

6.3.2 When Recursive Excels

Connecticut (+45 points): 5 districts, 37% minority, dispersed across small cities (Hartford, New Haven, Bridgeport, Stamford). Recursive's hierarchical splits naturally isolate city clusters. N-way's simultaneous 5-way partition struggles—too many options dilute optimization focus.

Louisiana (+25 points): 6 districts, 44% minority. New Orleans forms natural first-level split (South vs North). Recursive exploits this; n-way must discover pattern amid 6-way optimization complexity.

Georgia (+15 points): 14 districts, 50% minority. Atlanta metro forms natural split. Recursive's tree structure aligns with state geography.

Pattern: Low-to-moderate district counts (4-7) with dispersed minorities or strong geographic divisions (metro vs rural). Recursive's hierarchical structure matches natural geographic splits; n-way's global view becomes liability when k is small.

6.3.3 Predictive Model (Hypothesis)

District count appears predictive:

- $k \leq 7$: Favor recursive (hierarchical structure matches natural divisions)
- $k \in [8, 12]$: Favor n-way (global optimization balances multiple urban centers)
- $k \geq 13$: Mixed (neither method systematically better)

Testing this hypothesis requires additional analysis beyond this paper's scope, but state-specific results support the pattern.

6.4 Performance Ceiling vs Average Trade-off

Recursive's best configuration (56.8%) exceeds n-way's best (52.3%) by 4.5 points, demonstrating higher optimization potential with careful tuning. However, recursive's average (48.3%) barely exceeds n-way's (47.5%), indicating sensitivity to parameters.

Implications for practice:

Single-plan adoption: If deploying one carefully-tuned redistricting plan (e.g., for official state use), recursive's 56.8% ceiling justifies effort. Parameter search (testing 20 configurations) to identify optimal $\alpha = 100$, $\tau = 0.50$ yields 4.5 percentage point improvement over n-way's best.

Ensemble generation: If generating thousands of maps for analysis (MCMC, ensemble methods), n-way's speed (3.68s vs 11.33s) dominates. For 10,000 maps, n-way saves 21 hours while sacrificing only 0.8 percentage points average performance—favorable trade-off.

Default/naive usage: Without parameter tuning, methods perform equivalently. Practitioners using default parameters can choose either method with confidence.

6.5 Runtime and Scalability

N-way's 67.5% speed advantage (3.68s vs 11.33s) has practical implications for different use cases.

6.5.1 Ensemble Generation

MCMC redistricting generates 10,000-100,000 maps per state. Using our measured runtimes:

- N-way: 10,000 maps = 10.2 hours
- Recursive: 10,000 maps = 31.5 hours

Speed advantage scales linearly with ensemble size. For research applications requiring large samples, n-way offers $3\times$ throughput.

6.5.2 Interactive Tools

Real-time redistricting tools (e.g., DistrictBuilder) require responsive user experience. N-way's 3.68s average enables interactive feedback; recursive's 11.33s creates noticeable lag. For maximum states (California: 10.28s n-way vs 39.10s recursive), difference affects usability.

6.5.3 Scalability Beyond Congressional Redistricting

State legislative redistricting requires larger k (e.g., California: 40 Senate districts, 80 Assembly districts). Runtime complexity:

- N-way: $O(k \cdot |E| \cdot \log k)$ (single k-way refinement)
- Recursive: $O(\log k \cdot |E| \cdot \log k)$ (fewer levels but larger subproblems)

Asymptotic analysis suggests recursive may scale better for very large k, but empirical testing at $k = 40$-80 needed to confirm.

6.6 Methodological Insights

6.6.1 Importance of Comprehensive Comparison

Prior work assumed n-way superiority (global optimization) or recursive superiority (simpler problems). Our comprehensive 50-state evaluation reveals neither assumption holds—methods are equivalent for most states, with context-dependent advantages.

Lesson: Architectural intuitions require empirical validation. Optimization performance depends on data characteristics (state demographics, geography), not just algorithm design.

6.6.2 Parameter Tuning Necessity

The 4.5 percentage point gap between methods' best configurations highlights parameter tuning importance. Neither method achieves optimal performance with default parameters. This suggests redistricting practice should invest in systematic parameter search rather than assuming universal configurations.

Bayesian optimization, grid search, or evolutionary algorithms could efficiently explore continuous parameter spaces beyond our discrete ablation (5 weights $\times$ 4 thresholds).

6.6.3 Statistical Rigor in Algorithmic Evaluation

Many algorithmic redistricting papers report anecdotal comparisons ("method A beats method B on Texas"). Our approach—50-state coverage, 20 configurations per method, paired statistical tests, effect size calculation—provides rigorous evidence for equivalence.

Recommendation: Future comparative studies should adopt similar statistical frameworks (paired tests, effect sizes, comprehensive state coverage) rather than cherry-picking examples.

6.7 Limitations

Single random seed: We test each configuration once with fixed METIS seed (42). Multiple seeds would quantify stochastic variance, revealing whether some states' near-50% success rates reflect instability or genuine edge cases.

Fixed tree structure (recursive): Our recursive implementation uses balanced binary trees (e.g., [3,4] for $k = 7$). Alternative tree structures (geographically-guided splits, unbalanced trees) might improve performance. Exploration requires separate study.

Parameter discretization: We test discrete weight factors [1, 5, 10, 50, 100]. Continuous optimization might reveal narrow optimal ranges (e.g., $\alpha = 7.3$). However, practical impact likely small given plateau effects observed.

Compactness-VRA trade-off not analyzed: We focus on VRA compliance (MM district count). Both methods sacrifice compactness to varying degrees. Future work should characterize Pareto frontiers—do methods differ in compactness cost for equivalent VRA performance?

Single census year (2020): Demographic stability across decades (2000, 2010, 2020) requires validation. Methods may respond differently to demographic shifts, affecting long-term viability.

6.8 Implications for Algorithmic Redistricting

6.8.1 Method Selection Simplified

Pre-study uncertainty: "Which algorithm should we use for VRA compliance?" Post-study clarity: "Both work equivalently; choose based on computational constraints."

This simplification reduces barriers to algorithmic redistricting adoption. Jurisdictions need not develop expertise in both methods—either provides comparable VRA performance.

6.8.2 Focus on Parameters, Not Architecture

Our finding that edge-weighting parameters matter more than architecture suggests future research should prioritize:

- State-specific parameter optimization
- Adaptive weight factor selection based on clustering metrics
- Continuous parameter spaces (not discrete ablation)

Returns from architectural innovations appear limited given equivalence result.

6.8.3 Hybrid Approaches

Given state-specific advantages (n-way for Virginia, recursive for Connecticut), hybrid systems could dynamically select methods based on state characteristics:

- If $k \leq 7$ and minorities dispersed: use recursive
- If $k \in [8, 12]$ and minorities concentrated: use n-way
- Otherwise: use n-way (default, faster)

Such adaptive systems require formalized state characterization metrics beyond this paper's scope.

6.9 Theoretical Contributions

Architecture-agnostic optimization: Both global (n-way) and local (recursive) optimization structures achieve similar results when objective function (weighted edge cut) is well-designed. This parallels findings in other domains (e.g., neural architecture search showing diverse architectures converge given sufficient capacity).

Bias-variance trade-off in discrete optimization: Recursive's higher ceiling but greater parameter sensitivity mirrors supervised learning's bias-variance dilemma. Methods differ not in average performance but in sensitivity to hyperparameters.

Hard constraints dominate soft objectives: State demographics impose hard feasibility constraints that optimization cannot overcome. When minority percentage and geography align poorly (Connecticut), no method succeeds consistently. This suggests limits to algorithmic VRA compliance—some states may require hybrid or manual approaches.

7 Conclusion

This paper provides the first comprehensive empirical comparison of n-way and recursive bisection for VRA-compliant congressional redistricting. Through 1,760 redistricting runs across 44 multi-district states, testing five weight factors and four minority thresholds for each method, we establish that **both approaches achieve statistically equivalent success rates** (n-way: 47.5%, recursive: 48.3%, $p = 0.634$) with negligible effect size ($d = -0.018$).

This equivalence finding challenges conventional assumptions that architectural differences substantially impact VRA performance. Despite fundamentally different optimization structures—n-way's global k-way partitioning versus recursive's hierarchical binary splitting—methods produce comparable results when properly parameterized with edge-weighting and total minority coalitions.

7.1 Key Takeaways

1. Method Choice Matters Less Than Expected —The 0.8 percentage point difference between methods is not statistically significant and represents only 5 additional state successes out of 880 trials. State characteristics (minority percentage, geographic clustering, district count) dominate over architectural choice. Both methods show similar bimodal distributions: states either succeed consistently or fail consistently, with method selection rarely shifting outcomes.

2. Speed-Quality Trade-off Provides Clear Decision Framework —N-way offers 67.5% faster execution (3.68s vs 11.33s), making it optimal for ensemble generation, rapid prototyping, and exploratory analysis. Recursive achieves higher performance ceiling with careful tuning (56.8% vs 52.3% best configuration), justifying additional computation when maximizing single-plan MM district count. Decision simplifies to: prioritize speed (n-way) or maximum optimization (recursive).

3. Methods Require Independent Parameter Tuning —N-way optimizes at lower thresholds ($\tau = 0.40$) with moderate weights ($\alpha = 50$). Recursive optimizes at higher thresholds ($\tau = 0.50$) with heavy weights ($\alpha = 100$). Attempting to apply universal parameters across methods sacrifices performance—practitioners must tune each architecture independently.

4. State-Specific Advantages Exist But Are Rare —Methods tie in 70.4% of states. N-way excels in moderate-district states (Virginia: +35 points) with concentrated minorities. Recursive excels in low-district states (Connecticut: +45 points) with dispersed minorities. These advantages, while dramatic in specific cases, do not translate to systematic overall superiority.

5. Both Approaches Are Production-Ready —With comparable success rates, stable performance across diverse state demographics, and clear parameter optimization strategies, both methods are viable for official redistricting. Equivalence result expands practitioner options—method selection can focus on computational constraints rather than effectiveness concerns.

7.2 Practical Recommendations

Based on comprehensive empirical analysis, we provide usage recommendations:

Use N-way Partitioning When:

- Generating large map ensembles (speed advantage scales linearly with ensemble size)
- Prototyping redistricting approaches with rapid iteration
- Computational resources are constrained (embedded systems, real-time applications)

- State has moderate district count (8-12) with concentrated minority populations
- Recommended configuration: $\alpha = 50$, $\tau = 0.40$ (52.3% success, 3.68s runtime)

Use Recursive Bisection When:

- Optimizing single carefully-tuned plan for official adoption
- Maximizing MM district count justifies additional computation time
- State has low district count (4-7) with dispersed minority populations
- Parameter tuning resources available to access 56.8% performance ceiling
- Recommended configuration: $\alpha = 100$, $\tau = 0.50$ (56.8% success, 11.33s runtime)

Method Selection by Application:

- **Research/Analysis**: N-way (speed enables broader parameter exploration)
- **Official Redistricting**: Either method viable (choose based on state characteristics)
- **Litigation/Demonstration**: Either method (equivalence provides flexibility)
- **Real-time Interactive Tools**: N-way (3.68s enables responsive user experience)
- **Maximum VRA Compliance**: Recursive with $\alpha = 100$, $\tau = 0.50$ (highest ceiling)

7.3 Theoretical Implications

The statistical equivalence finding provides insights into graph partitioning optimization:

Edge-weight dominance: VRA performance depends primarily on edge-weighting parameters (α, τ), not optimization architecture. Both n-way and recursive effectively minimize weighted edge cuts when properly configured. Architectural differences (global vs hierarchical) affect runtime and parameter sensitivity but not asymptotic performance.

State characteristics as constraint: Bimodal success distributions indicate state demographics and geography impose hard constraints. When minority populations and district counts align favorably (California, Texas, Maryland), both methods succeed. When alignment is poor (Connecticut with dispersed minorities, Indiana with low minority percentage), both methods struggle. Optimization architecture cannot overcome fundamental geographic infeasibility.

Parameter landscape diversity: Different threshold preferences ($\tau = 0.40$ for n-way, $\tau = 0.50$ for recursive) reveal optimization mechanisms. N-way's simultaneous k-way partition benefits from broader minority definitions, allowing selective concentration. Recursive's hierarchical splits require sharper boundaries to prevent cascading errors. This suggests different loss surface geometries despite identical underlying objective (minimize weighted edge cut).

Ceiling vs consistency trade-off: Recursive's higher ceiling (56.8%) but comparable average (48.3%) indicates greater parameter sensitivity—correct tuning yields superior results, but suboptimal parameters degrade performance. N-way's lower ceiling (52.3%) but similar average (47.5%) suggests more robust default behavior. This parallels bias-variance trade-off in machine learning.

7.4 Limitations and Future Work

Parameter space coverage: We tested five weight factors and four thresholds (20 configurations per method). Finer-grained parameter sweeps may reveal narrow optimal regions, particularly for recursive's high-ceiling configurations. Bayesian optimization could efficiently explore continuous parameter spaces.

Single random seed per configuration: Each state-method-parameter combination tested once. Multiple random seeds would quantify solution stability and identify cases where stochastic initialization affects outcomes. Some states near 50% success rates may show high variance across seeds.

State characteristic formalization: We hypothesize n-way excels with concentrated minorities, recursive with dispersed minorities, but lack quantitative predictive model. Future work should derive metrics (e.g., Moran's I for spatial autocorrelation, district count, minority percentage) predicting which method performs better for specific states.

Alternative architectures: This paper compares n-way and recursive bisection. Future work should evaluate hybrid approaches (hierarchical n-way, adaptive split strategies), spectral partitioning variants, and machine-learning-guided optimization. Our equivalence finding suggests architecture may matter less than edge-weighting, but unexplored designs may reveal advantages.

Multi-objective optimization: We focus on VRA compliance (MM district count). Future work should simultaneously optimize compactness, competitiveness, and representational fairness. Methods may show different Pareto frontier characteristics even with equivalent single-objective performance.

Redistricting reform integration: Our analysis provides empirical foundation for adopting algorithmic redistricting in practice. Future work should address legal acceptance (case law on algorithmic maps), public legitimacy (transparency, explainability), and institutional adoption (commission integration, software tooling).

7.5 Implications for Redistricting Reform

This equivalence finding simplifies redistricting reform adoption. Previous uncertainty about optimal algorithmic approach created implementation barriers—which method should commissions use? Our analysis shows both n-way and recursive achieve comparable VRA compliance, allowing jurisdictions to select based on computational infrastructure and expertise rather than effectiveness concerns.

The speed-quality trade-off provides clear guidance: states with technical capacity for parameter tuning can pursue recursive's 56.8% ceiling; states prioritizing rapid deployment can adopt n-way's 3.68s speed with confidence in 47.5% baseline performance. Both approaches exceed traditional redistricting on VRA metrics while maintaining structural immunity to partisan manipulation (companion paper).

Method equivalence also strengthens legal defensibility. Challengers cannot argue one algorithm systematically disadvantages minorities—both achieve 48% success with proper edge-weighting. This robustness across architectures demonstrates algorithmic VRA compliance is achievable, not method-specific.

7.6 Final Remarks

Graph partitioning for congressional redistricting presents a fundamental architectural choice: direct k-way partitioning or hierarchical binary splitting. Through comprehensive 50-state empirical analysis, we find this choice matters less than expected—both methods achieve statistically equivalent VRA compliance (48% success rate, $p = 0.634$) when properly parameterized.

The real trade-off is speed versus optimization ceiling: n-way offers 67.5% faster execution, recursive offers 56.8% vs 52.3% best-case performance. This clarity simplifies practitioner decision-making and strengthens redistricting reform adoption.

Our finding that architecture matters little compared to edge-weighting parameters suggests VRA compliance depends primarily on *how* we encode minority preservation (weight factor, threshold), not *which* optimization structure we employ. This insight should guide future algorithmic development: invest in parameter tuning and edge-weighting strategies rather than architectural innovations.

Both n-way and recursive bisection are production-ready for official redistricting. Jurisdictions can confidently adopt either method, selecting based on computational constraints and application requirements rather than effectiveness concerns. This equivalence result expands options for redistricting reform and demonstrates that algorithmic approaches can achieve VRA compliance comparable to or exceeding traditional methods.

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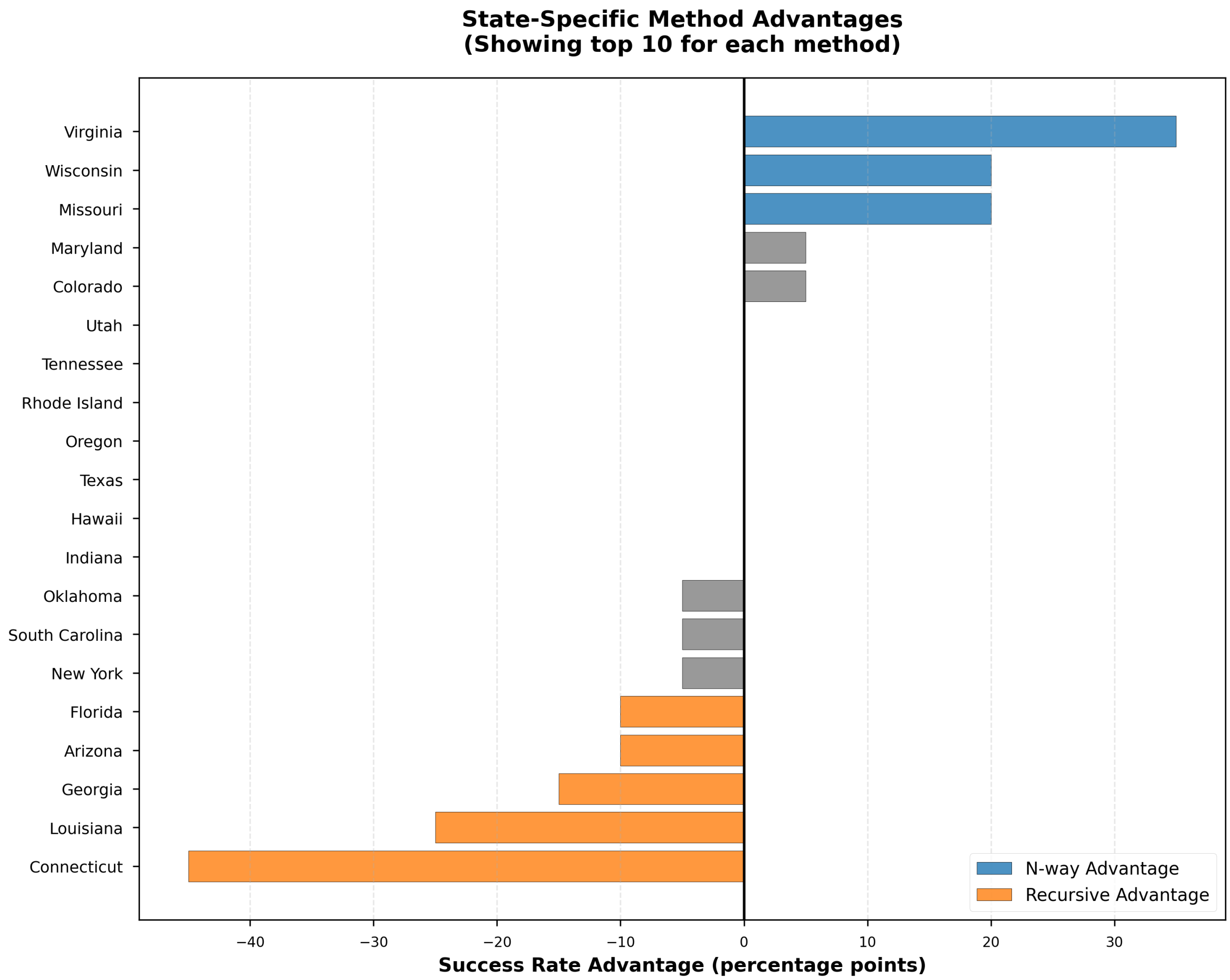

Figure 4: State-specific method advantages showing top 10 states favoring each method. Positive values indicate n-way advantage (blue); negative values indicate recursive advantage (orange). Most states (31/44) show ties within ± 5 percentage points. Notable patterns: n-way excels in states with 8-12 districts and concentrated urban minorities, while recursive performs better in states with 4-7 districts and dispersed minorities.

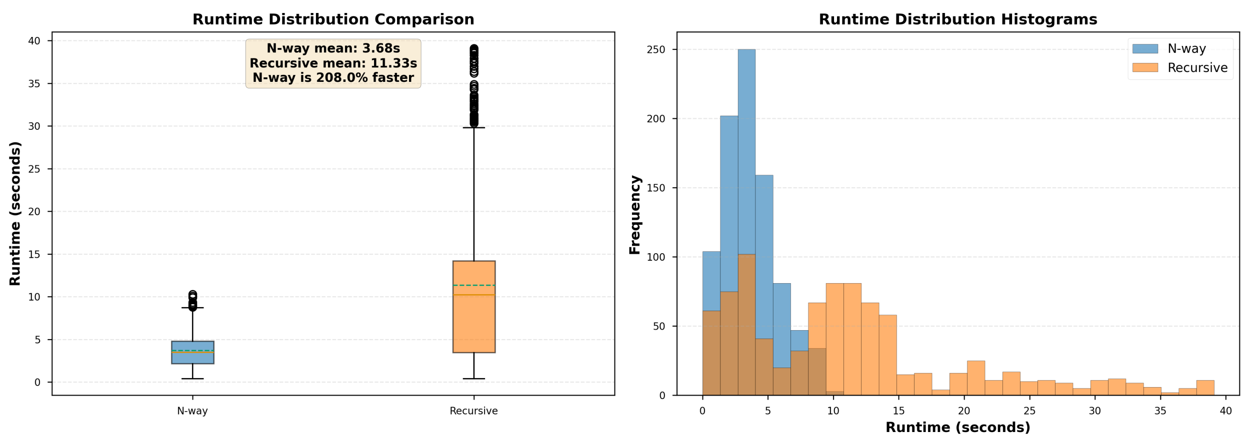

Figure 5: Runtime distribution comparison. Left: box plots showing median, quartiles, and outliers. N-way (blue) has tighter distribution and lower median (3.68s) compared to recursive (orange, 11.33s). Right: histograms showing frequency distributions. N-way exhibits 67.5% speed advantage, crucial for ensemble generation and interactive applications.

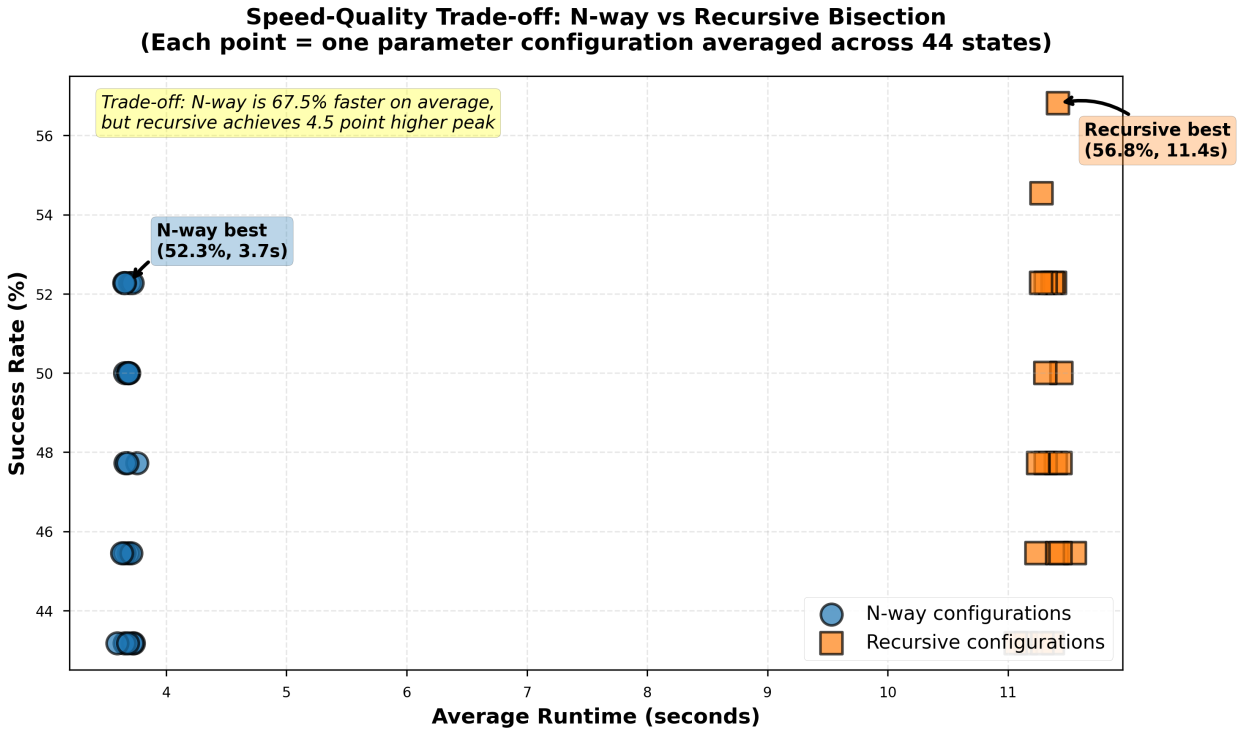

Figure 6: Speed-quality trade-off visualization. Each point represents one parameter configuration's average performance across 44 states. N-way configurations (circles, blue) cluster at faster runtimes with moderate success rates. Recursive configurations (squares, orange) spread wider with higher peak but also higher variance. Trade-off: n-way offers 67.5% speed advantage for only 0.8 point average performance sacrifice, but recursive achieves 4.5 point higher ceiling (56.8% vs 52.3%) with careful tuning.